

SAI NARAYAN SUNDARESAN

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RESEARCH INTERESTS

My research interests are in designing efficient systems for AI training and deployment, leveraging model architecture and exploiting computational redundancy across modalities to make generative systems scalable and efficient in real-world settings.

EDUCATION

- **Ph.D. in Computer Science**, Advisor: Prof. Chunwei Liu Aug 2026 – Present
Purdue University, West Lafayette, IN
- **Dual Degree (B.Tech, M.Tech), Industrial and Systems Engineering** Jul 2019 – Apr 2024
Indian Institute of Technology Kharagpur, India
Micro-specialization: Artificial Intelligence and Applications
GPA: 9.15/10.00

RESEARCH EXPERIENCE

- **Research Associate – Adobe Research (Systems and Insights Group)** Jun 2024 – Aug 2026
under Dr. Subrata Mitra, Dr. Atanu Sinha, Dr. Shiv Saini
 - * Devised an efficient no-decode source attribution framework to improve citation accuracy without decode overhead
 - * Worked on building efficient systems for LLM serving by implementing a prefix-independent cache reuse strategy
 - * Developed a vLLM-based streaming attribution system by parallelizing server attribution requests with API calls
 - * Built a heterogeneity-aware clusterwise behavior prediction model to predict user trajectories from clickstream data
- **Research Intern – Sarvam.AI** Jan 2024 – May 2024
under Dr. Pratyush Kumar, Dr. Vivek Raghavan
 - * Built an end-to-end teaching tool that creates a guided audio lesson from text using custom speech-to-text pipelines
 - * Deployed an efficient speech recognition service for production using NVIDIA's RIVA toolkit for low-resource languages
 - * Trained expressive TTS models with a StyleTTS2-based architecture spanning diverse emotions and speech patterns
- **Research Intern – Adobe Research** May 2023 – Aug 2023
under Dr. Srikrishna Karnam
 - * Built a design creation tool for making posters given a product image and brand reference based on Custom Diffusion
 - * Devised an algorithm to determine the optimal latents for multiple reference concepts based on latent interpolation
 - * Implemented the algorithm in Custom Diffusion to reduce the dependence on the initial latent in the diffusion process
- **Research Intern – AI4Bharat, IIT Madras** Jan 2023 – Apr 2023
under Prof. Mitesh M. Khapra
 - * Collated benchmarks for ASR across 12 Indian languages (10k hrs), accounting for domain and quality variations
 - * Fine-tuned OpenAI's Whisper model on the dataset, achieving an average 4.1 WER reduction over existing solutions

PUBLICATIONS

- [1] ([COLM'26](#)) Sai Sundaresan*, Archit Gupta*, et al. **Attr-Kit: An Efficient Toolkit for No-Decode Source Attribution**. In *Conference on Language Modelling*, 2026.
- [2] ([EMNLP'25](#)) Sai Sundaresan, et al. **Subjective Behaviors and Preferences in LLM: Language of Browsing**. In *The 30th Conference on Empirical Methods in Natural Language Processing*, 2025.
- [3] ([SIGMOD'25](#)) Shubham Agarwal*, Sai Sundaresan*, et al. **Cache-Craft: Managing Chunk-Caches for Efficient Retrieval-Augmented Generation**. In *The 44th International Conference on Management of Data*, 2025.
- [4] ([ICLST'24](#)) Sai Sundaresan, Anand Abraham. **Single Resource Capacity Control Model for Hidden City Ticketing**. In *The International Conference on Logistics, Supply Chain and Transportation*, 2024.
- [5] ([ORSI'24](#)) Sai Sundaresan, Anand Abraham. **Clearance Sale Models under Competition**. In *The International Conference on Trends in Business Analytics & Management*, 2024.
- [6] ([INTERSPEECH'23](#)) Kaushal Santosh Bhogale*, Sai Sundaresan*, et al. **Vistaar: Diverse Benchmarks and Training Sets for Indian Language ASR**. In *The 24th INTERSPEECH Conference*, 2023.

ACTIVITIES AND ACHIEVEMENTS

- Received best paper award and merit paper award, respectively, for work presented at ICLST '24 and ORSI '24
- Contributed to multiple paper reviews for technical conferences including USENIX ATC, SIGMOD, and EMNLP
- Mentored over 6 undergraduate interns and collaborated with 1 PhD intern during summer internships at Adobe
- Received Certificate of Merit for achieving the top rank in Computer Science, AISSCE 2019 (D.A.V. School, Chennai)